ICT2217

Network Security

SINGAPORE INSTITUTE OF TECHNOLOGY

Lab 1.2 Notes:

Internal Attacks over LAN and Switch Security Configuration I

Lab Exercise 1.2

1 Switch spoofing attack

2 Defenses:

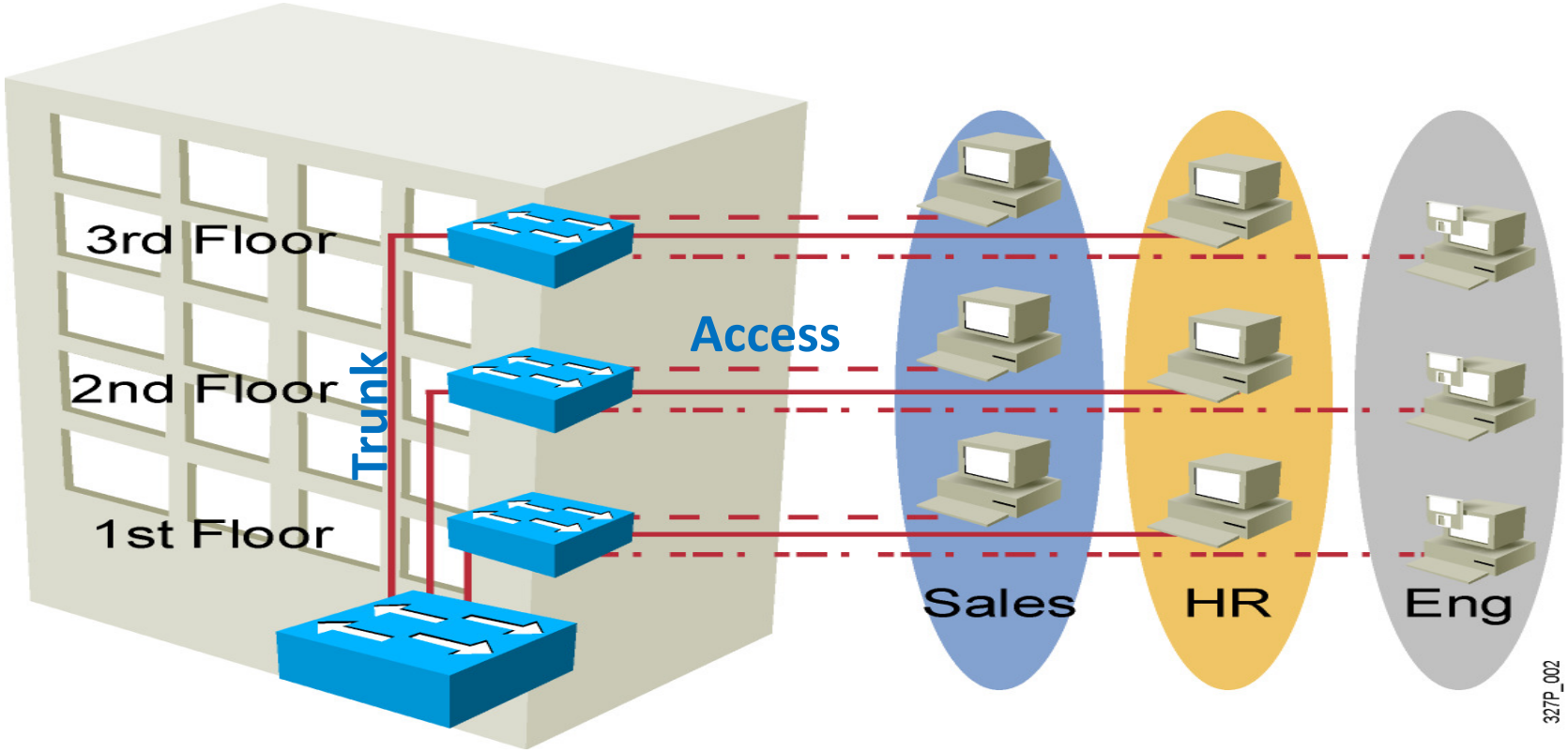
- Manually configure ports as access or trunk
- Disable DTP
- Good practice to shut down all unused ports and configure them into an unused VLAN

In ICT1013, you have learned about configuring VLANs (virtual LANs) on switches to separate frames in different VLANs to support users in different departments.

With VLANs, switch port connected to a link is operating either as:

(i) Access – carry frames for only one VLAN configured on the port; or

(ii) Trunk – carry frames for multiple VLANs on one physical link



To simplify configuration, modern switches are commonly shipped with mechanisms or protocols to automate trunk formation; e.g. the Cisco Dynamic Trunking Protocol (DTP).

In Cisco switches, each switch port can be configured in 4 different modes: access, dynamic auto, trunk and dynamic desirable.

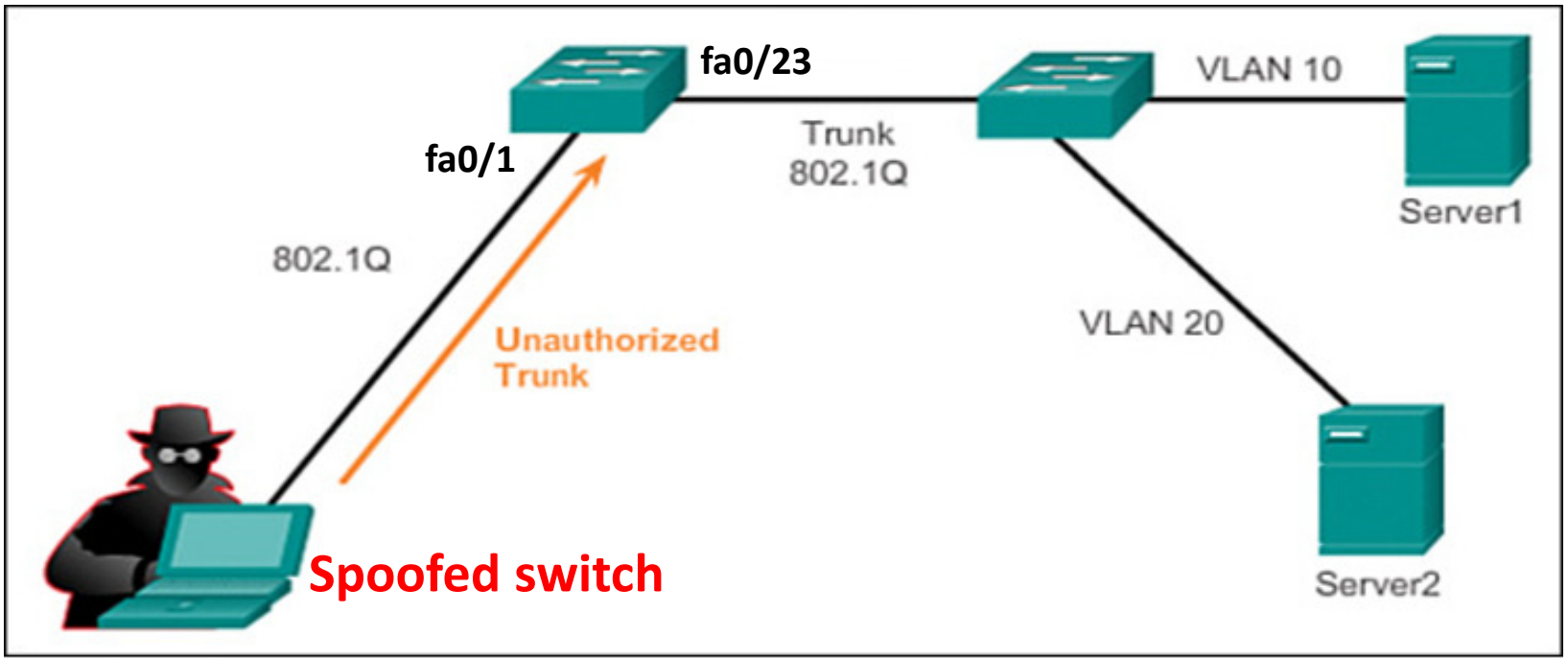
Then, DTP will be run automatically which will result in the link being an access or trunk as follows:

Administrative Mode	Access	Dynamic Auto	Trunk	Dynamic Desirable
access	Access	Access	Do Not Use[1]	Access
dynamic auto	Access	Access	Trunk	Trunk
trunk	Do Not Use[1]	Trunk	Trunk	Trunk
dynamic desirable	Access	Trunk	Trunk	Trunk

By default, Cisco switch ports are in dynamic auto mode which enables DTP and will passively wait to receive trunk negotiation to be come a trunk.

Unfortunately, the use of DTP can lead to switch spoofing attack – a form of VLAN hopping attack where attackers could access frames in other VLANs that normally would not be accessible.

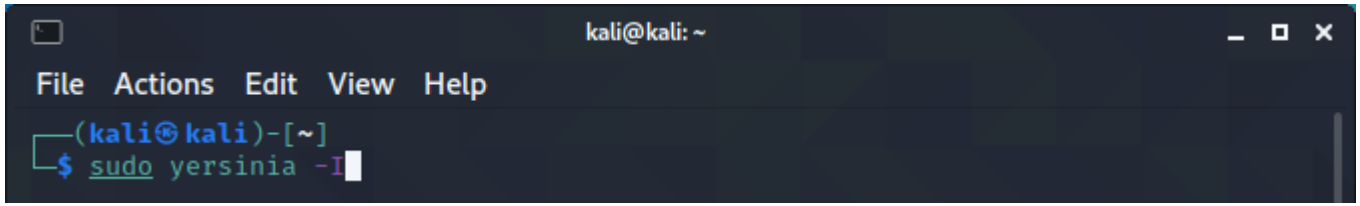
Taking advantage of the default configuration of the switch port which is dynamic auto, an attacker can inject DTP messages to form a trunk.

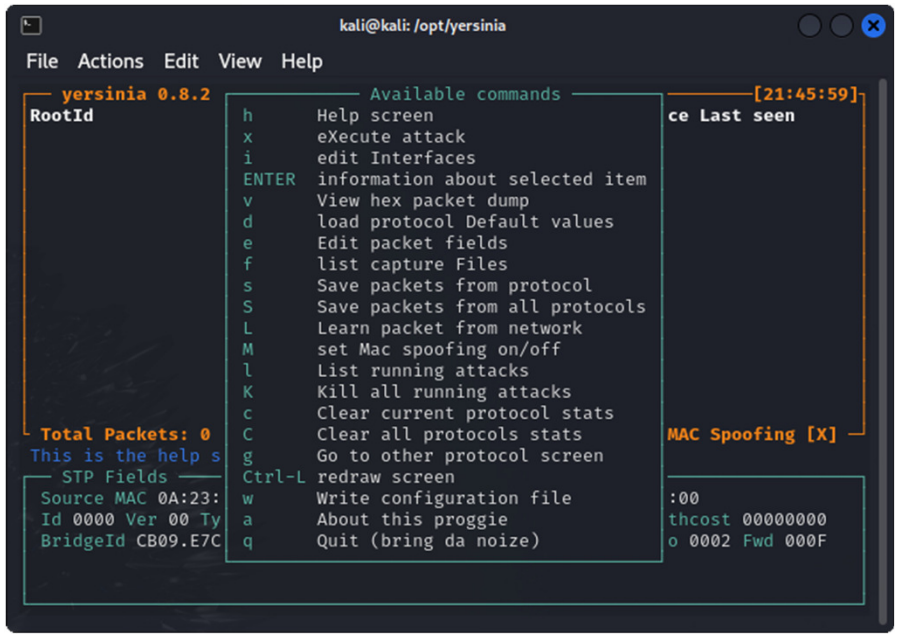


With the spoofed switch, the attacker may now receive frames from other VLANs, or send spoofed frames tagged for any target VLAN.

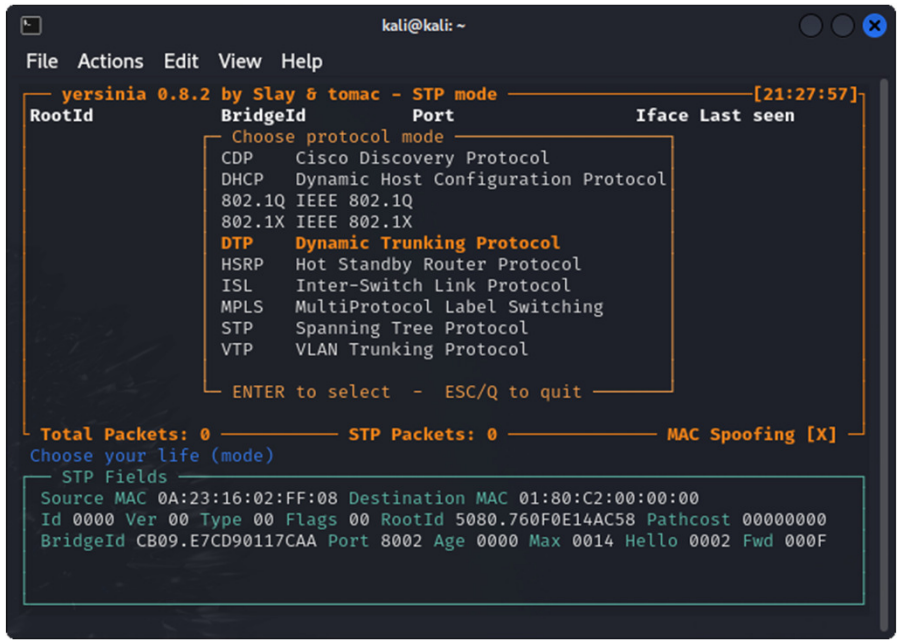
To demonstrate switch spoofing attack, you may use the tool Yersinia, which is available in Kali Linux.

- Launching Yersinia in interactive mode:





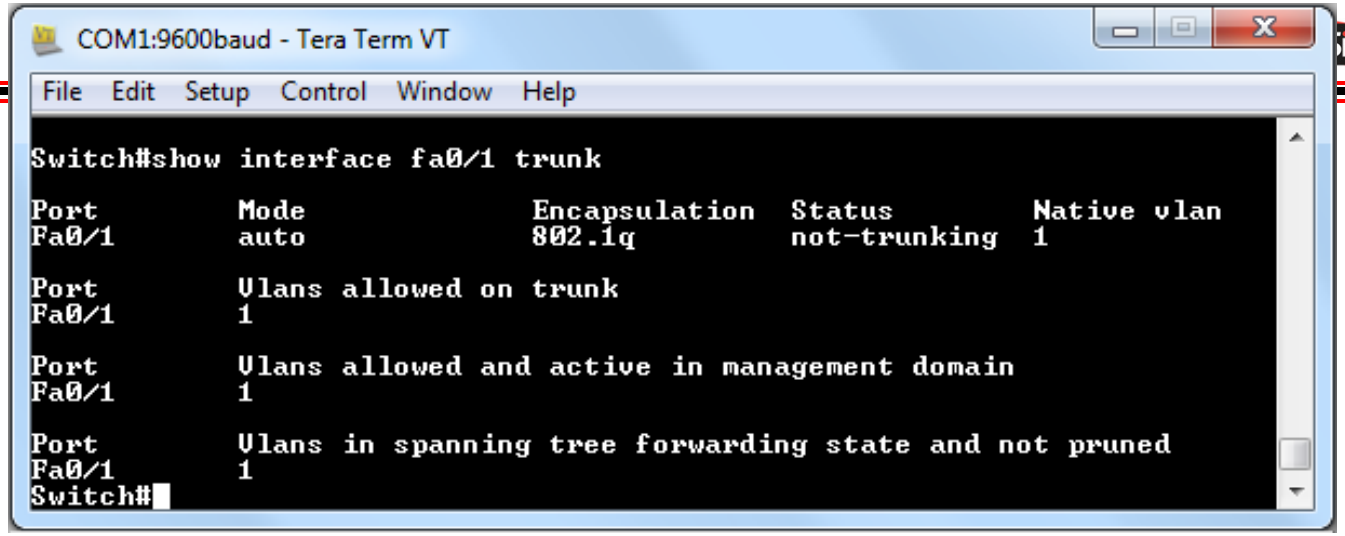
Press key ‘h’ to display help screen on the functionality of different keys.



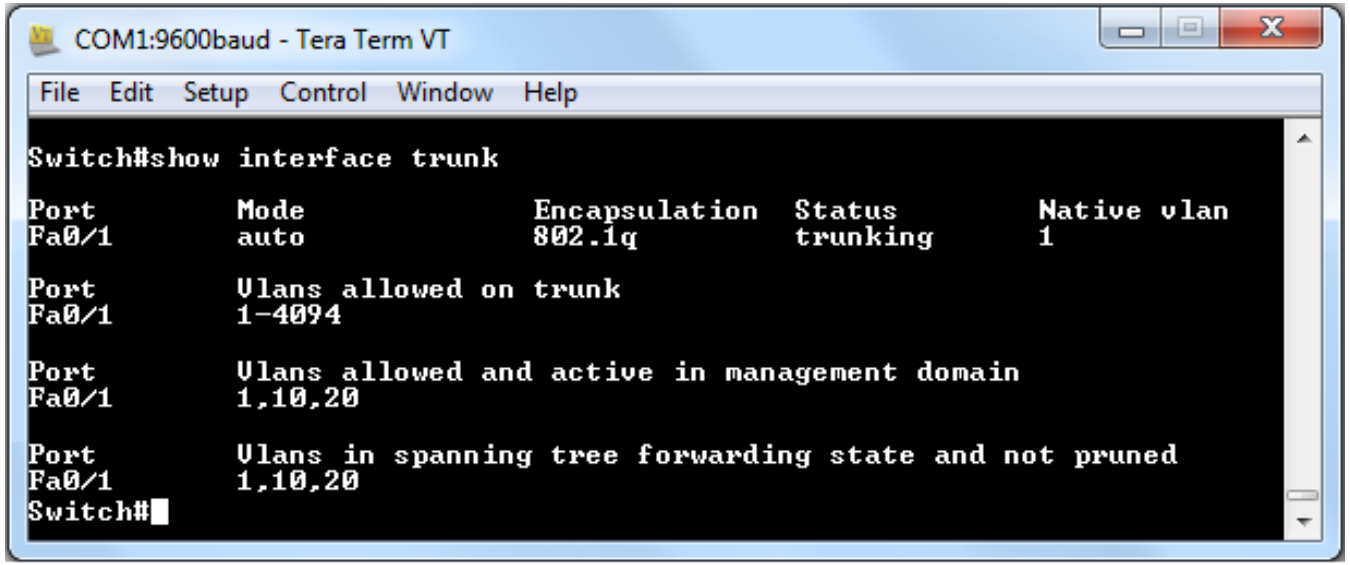
Press key ‘g’ to display protocol screen, arrow keys to navigate and ENTER key to select.

An example of the result of conducting switch spoofing attack on a switch using Yersinia.

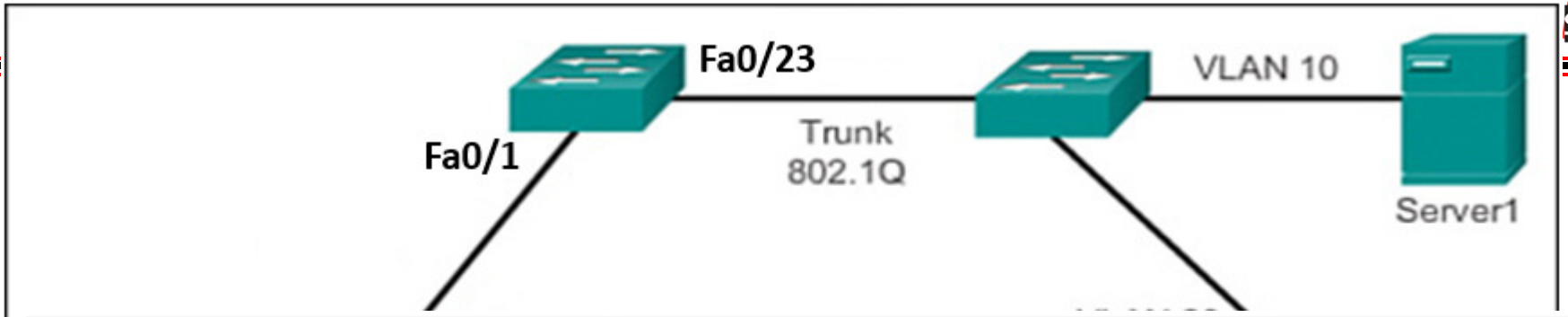
Port Fa0/1 in default dynamic auto before attack:



Port Fa0/1 becomes trunk after attack:



To prevent switch spoofing attack, disable DTP negotiations and manually configure ports as access or trunk explicitly.

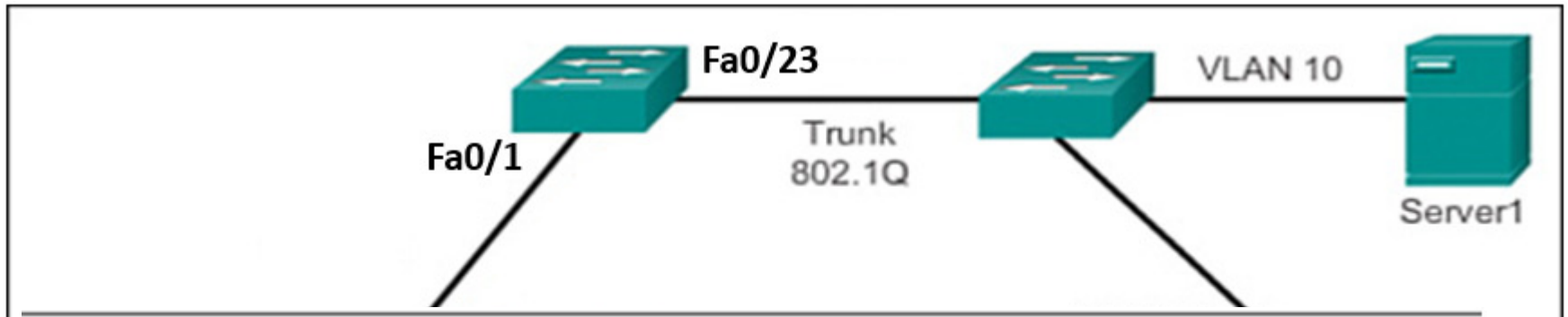



```
SW2(config)# interface fa0/1
! Specifies that this is an access port
SW2(config-if)# switchport mode access
SW2(config-if)# switchport access VLAN 10
! Note, even for an interface configured as an access port,
! negotiation packets are still being sent unless we disable it
SW2(config-if)# switchport nonegotiate

SW2(config-if)# interface fa0/23
! Specifies the port as a trunk
SW2(config-if)# switchport mode trunk
SW2(config-if)# switchport nonegotiate
```

For added security, it is also wise to shutdown all unused ports and place them in an unused VLAN , e.g. 100, so as to prevent someone from discovering a live port that might be exploited.

For example, if port Fa0/2-22 are unused, then shut them down to prevent attacker from discovering and exploiting.




```
SW2(config)# interface range fa0/2-22
SW2(config-if)# switchport access VLAN 100
SW2(config-if)# shutdown


SW2(config-if)# interface fa0/23
SW2(config-if)# switchport trunk allowed vlan remove 100
```

Prevent trunk from carrying unused VLAN 100 frames